GRAPH THEORY:
HOMEWORK 8
BOOK SECTION 4.1 PART II AND 4.2.

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Theorem 4.1.11 if G is a 3-regular graph then $\kappa(G) = \kappa'(G)$.

Proposition 4.1.12 If S is a set of vertices in a graph G, then

$$\left|[S, \overline{S}]\right| = \Big[\sum_{v \in S} d(v)\Big] - 2e(G[S])$$

Corollary 4.1.13 If G is a simple graph and $|[S, \overline{S}]| < \delta(G)$ for some nonempty proper subset $S \subset V(G)$, then $|S| > \delta(G)$.

1. Let G be a connected graph in which for every edge e, there are cycles C_1 and C_2 containing e whose only common edge is e. Prove that G is 3-edge-connected. Use this to show that the Petersen graph is 3-edge-connected

Proof. Suppose an edge cut F has size at most two, and choose an edge $e \in F$. Let S be a component of $G - F$ containing one endpoint of e. (We may take F to be a minimal disconnecting set, so its edges join distinct components.) Every cycle containing e must cross from S to $\overline{S}$ again, and hence must contain another edge of F. Thus, if $|F| = 1$, no such cycle exists; if $|F| = 2$, both C_1 and C_2 contain the other edge of F. Both conclusions contradict the hypothesis. Therefore $\kappa'(G) \geq 3$.

Label the Petersen graph with vertices a_i, b_i for $i \in \mathbb{Z}/5\mathbb{Z}$, with edges a_ia_{i+1}, a_ib_i, and b_ib_{i+2}. Rotating the subscripts preserves adjacency, so it suffices to check one edge of each type. The following pairs of cycles have only the indicated edge in common:

edge	C_1	C_2
a_0a_1	$a_0a_1a_2a_3a_4a_0$	$a_0a_1b_1b_3b_0a_0$
a_0b_0	$a_0b_0b_2a_2a_1a_0$	$a_0b_0b_3a_3a_4a_0$
b_0b_2	$b_0b_2b_4b_1b_3b_0$	$b_0b_2a_2a_1a_0b_0$

Hence the Petersen graph is 3-edge-connected. Since it is 3-regular, deleting the three edges at a vertex disconnects it, so its edge connectivity is exactly three. □

2. Use Proposition 4.1.12 and Theorem 4.1.11 to prove that the Petersen graph is 3-connected

Proof. Let G be the Petersen graph. It is simple, 3-regular, and has ten vertices and girth five. To see the girth assertion using the labeling above, a cycle with no spokes a_ib_i is a 5-cycle. A cycle with two spokes contains a nonempty outer path and a nonempty inner

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path. These paths cannot both have length one, since outer adjacency requires subscript difference ± 1 and inner adjacency requires difference ± 2. Such a cycle therefore has length at least five. A cycle with at least four spokes has at least four intervening nonspoke edges, and has length at least eight.

Suppose there were an edge cut $[S, \overline{S}]$ of size at most two, choosing S to be the smaller side. By Corollary 4.1.13, $|S| > 3$, and since $|V(G)| = 10$, we have $|S| \in \{4, 5\}$. If $|S| = 4$, the girth condition makes $G[S]$ a forest, so $e(G[S]) \leq 3$. If $|S| = 5$, either $G[S]$ is a forest or any cycle in it uses all five vertices. That 5-cycle has no chord, since a chord would create a triangle. Thus $e(G[S]) \leq 5$. Proposition 4.1.12 now gives

$$|[S, \overline{S}]| = 3|S| - 2e(G[S]) \geq \begin{cases} 12 - 6 = 6, & |S| = 4, \\ 15 - 10 = 5, & |S| = 5, \end{cases}$$

a contradiction. Therefore $\kappa'(G) \geq 3$, and the degree bound gives $\kappa'(G) = 3$. By Theorem 4.1.11, $\kappa(G) = \kappa'(G) = 3$, so G is 3-connected. □

3. Let G be a simple graph with diameter 2 and let $[S, \overline{S}]$ be a minimum edge cut with $|S| \leq |\overline{S}|$.

(a) Prove that every vertex of S has a neighbor in $\overline{S}$.
(b) Use part (a) and Corollary 4.1.13 to prove that $\kappa'(G) = \delta(G)$.

Proof. Write $F = [S, \overline{S}]$. Since the edges incident with a minimum-degree vertex form an edge cut, $|F| = \kappa'(G) \leq \delta(G)$.

(a) Suppose some $x \in S$ has no neighbor in $\overline{S}$. For every $y \in \overline{S}$, the diameter-two hypothesis supplies a path x, z, y of length two. Since every neighbor of x lies in S, the edge zy belongs to F. Consequently each vertex of $\overline{S}$ is incident with an edge of F, and these give distinct edges for distinct y. Hence

$$|F| \geq |\overline{S}| \geq |S| > d(x) \geq \delta(G),$$

where $d(x) \leq |S| - 1$ because G is simple. This contradicts $|F| \leq \delta(G)$. Thus every vertex of S has a neighbor in $\overline{S}$.

(b) By part (a), $|F| \geq |S|$. If $\kappa'(G) = |F| < \delta(G)$, Corollary 4.1.13 gives $|S| > \delta(G)$, yielding $|F| \geq |S| > \delta(G)$, a contradiction. Therefore $\kappa'(G) = \delta(G)$. □

4. Prove or disprove: If P is a u, v-path in a 2-connected graph G, then there is a u, v-path Q that is internally disjoint from P.

Proof. The assertion is false. Let G be the 4-cycle with cyclic vertex order u, a, v, b, u, together with the diagonal ab. The spanning 4-cycle shows that deleting any one vertex leaves a connected graph, so G is 2-connected. However, $P = u, a, b, v$ is a u, v-path whose internal vertices are a and b. A u, v-path internally disjoint from P would have to use only u and v, and hence would have to be the edge uv. That edge is absent. Thus no such Q exists. □

5. Prove that a graph G with at least four vertices is 2-connected if and only if for every pair X, Y of disjoint vertex sets with $|X|, |Y| \geq 2$, there exist two completely disjoint paths P_1 and P_2 in G such that each has an endpoint in X and an end point in Y and no internal vertex in X or Y.

Proof. First suppose G is 2-connected, and let X, Y be disjoint with $|X|, |Y| \geq 2$. Add two new vertices s, t, joining s to every vertex of X and t to every vertex of Y. No single vertex other than s, t separates s from t: after deleting any vertex w of G, the graph $G - w$ is connected and both $X \setminus \{w\}$ and $Y \setminus \{w\}$ remain nonempty. By the vertex version of Menger's theorem there are two internally vertex-disjoint s, t-paths. Removing s, t leaves two vertex-disjoint paths from X to Y. On each, retain the segment ending at its first vertex in Y and starting at the last vertex in X before that endpoint. These segments remain vertex-disjoint and have no internal vertex in $X \cup Y$, as required.

Conversely, assume the stated path property. First G is connected. Otherwise, if some component is a singleton a, choose distinct other vertices b, c, d and put $X = \{a, b\}$, $Y = \{c, d\}$. Two vertex-disjoint X, Y-paths would have to use both vertices of X as endpoints, but no path can join a to Y. If there is no singleton component, choose two vertices of one component as X and two vertices of another component as Y; there is then no X, Y-path at all. Both possibilities contradict the property.

Suppose now that z is a cut vertex. If $G - z$ has a singleton component a, at least two other vertices b, c remain because $|V(G)| \geq 4$. Take $X = \{a, z\}$ and $Y = \{b, c\}$. The path starting at a must pass through z, while the other path must have z as its endpoint, contradicting vertex disjointness. If $G - z$ has no singleton component, choose two vertices of one component as X and two of another as Y. Every X, Y-path passes through z, so two such paths cannot be vertex-disjoint. Thus G has no cut vertex. Being connected with at least four vertices, it is 2-connected. □